

Criteria A

Inquiring and Analysing

Designers do not start by making things.



First, they:

- **Understand the problem**
- **Learn about the users**
- **Find out what is needed**

Then they create a solution



What is Criteria A?

Criteria A = Inquiring and Analysing

Designers ask questions before they design.

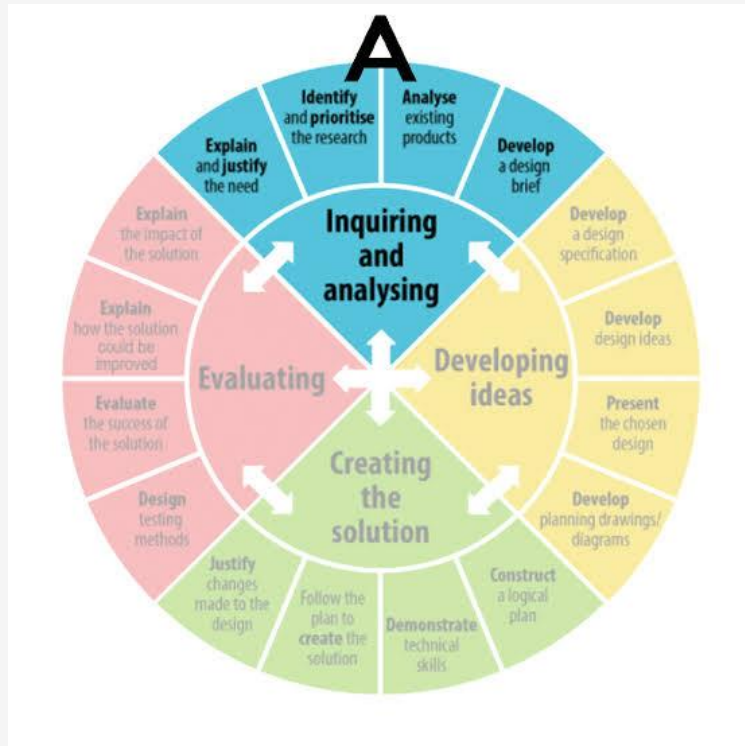
They:

- ✓ Understand the problem
- ✓ Learn about users
- ✓ Research information
- ✓ Study existing products
- ✓ Decide what is important



Criteria A: Inquiring and Analysing

- Inquire → Develop Ideas → Create → Evaluate



Strand i: Explain and justify the need for a solution to a problem

Command Terms:

Explain

- Give reasons and details.

Justify

- Give reasons that support your answer.

Question:

Why are traffic lights important?

Weak answer:

Traffic lights are useful.

Better answer:

Traffic lights are important because they help control traffic and reduce accidents.

Explain = What + Why

Example:

Why should students wear helmets while cycling?

Justification:

Students should wear helmets because they protect the head and reduce injuries during accidents.

Justify = Give evidence and reasons



Explain Vs Justify

Explain	Justify
Tell why	Prove why
Give details	Give reasons and evidence
Describe importance	Convince the reader

Example:

- **Explain:**
Interactive games help students learn through engagement.
- **Justify:**
Interactive games should be used because they increase participation and provide immediate feedback.



Explain and Justify the Need for a Solution

Today we will answer:

- Why is this problem important?
- Who are the users?
- How will the solution help?
- Why does the solution need to exist?



Design Situation

Young learners use many digital systems every day such as games, apps and learning platforms. These systems communicate with users through inputs, outputs and interactive responses.

In this unit, students will explore how simple block-based coding can be used to design interactive digital solutions. Through the design cycle, they will investigate user needs, develop ideas, create a simple interactive project and evaluate its effectiveness.

Students will create a interactive project using Pictoblox that demonstrates communication between a user and a digital system.



Our Design situation is...

Young learners use:

- Games
 - Apps
 - Learning platforms
- every day.

These systems communicate using:

INPUT → PROCESS → OUTPUT



Task Introduction

Now you will begin Strand i.

You need to:

- ✓ Explain why interactive systems are important
- ✓ Identify users
- ✓ Explain how your solution helps users
- ✓ Justify why your solution is needed



Summative Task (Strand i)

Question 1

- Why are interactive digital systems important?
Think: Where do we use them? And How do they help us?
- Who will use your project?
Think: Classmates? /Younger students? /Teachers?
- How will your project help or engage users?
Think: Teach? /Entertain? /Give information?
- Why does your solution need to exist?
Think: What problem does it solve? /What benefit does it provide?



a) Explain why interactive digital systems are important in everyday life.

b) Identify the users of your proposed interactive digital solution.

c) Explain how your solution could help, engage or communicate with users.

d) Justify why there is a need for your solution.



Strand ii: State and prioritize the main points of research needed to develop a solution to the problem.

Command Terms

State

Give a short answer without explanation.

Example:

Question: What is your favourite game?

State:

Minecraft

Prioritize

Arrange items from most important to least important.

Example:

Creating a School Event

Choose a date

Book a venue

Create posters

Decorate



Strand ii: State and prioritize the main points of research needed to develop a solution to the problem.

"Before making a game, what information would you need to know?"

You need to know:

- What users like
- What makes a game fun
- How PictoBlox works
- How scores are added

Designers research before they create.



Why Do Designers Research?

Imagine creating a game without research.

- ✗ Users do not enjoy it
- ✗ Instructions are confusing
- ✗ The game does not work properly

Research helps us make better decisions.



What Research Means in This Unit

We are creating an interactive digital solution.

We need to research on:

- Users
- Features
- Coding blocks
- Existing games



What Could We Research? (Stating)

I am going to find out:

- ✓ What features make an interactive game interesting.
- ✓ What users expect from a digital game.
- ✓ Which PictoBlox blocks respond to user input.
- ✓ How scores or points can be tracked.
- ✓ What makes a game easy to use.



Which Research Point is Most Important? (Prioritize)

Suppose you want to make a game.

Which should come first?

- A. Fancy colours
- B. What users want

Most designers would choose:

What users want

- Because the solution is being made for users.



Summative Task Introduction

You need to:

- ✓ State four research points
- ✓ Prioritize them
- ✓ Explain why your highest-priority research point is important



Strand (ii) State and prioritize the main points of research needed to develop a solution to the problem

Research Planning: To develop an effective interactive digital solution, identify the information you need to research.

a) State at least four things you need to research before creating your solution.

Examples:

- I am going to find out what features make an interactive program interesting for users.
- I am going to interview my classmates to learn what they expect from an interactive digital game.
- I am going to research which coding blocks can be used to make a program respond to user input.
- I am going to search for examples of simple interactive digital projects created by beginners.
- I am going to research how scores, points, or attempts can be tracked in a program.
- I am going to investigate what makes a digital solution easy to use and understand.

b) Prioritize the research points from most important to least important.

Priority	Research Point
Example:	I am going to interview my classmates to learn what they expect from an interactive digital game.
1	
2	
3	
4	

c) Explain briefly why your highest-priority research point is important.

Strand iii: Describe the main features of one existing product that inspires the solution

Command Term: Describe

Describe: Give a detailed description about something.

Example:

- **Poor Answer:** "The game is fun."
- **Better Answer:** "The game asks math questions, gives immediate feedback and shows the player's score."

Describe = Give details.



Strand iii: Describe the main features of one existing product that inspires the solution

Before creating something new, designers look at existing products.

Know about the product:

- What does it do?
- What works well?
- What can be improved?
- What ideas can inspire me?

This is called **Product Analysis**.



Strand iii: Describe the main features of one existing game that inspires the solution

- **Describe** (Explain in detail)
- the **main features** (Important parts)
- of **one existing product** (Any existing game or any number guessing game)
- that **inspires** (Provides ideas, not copying)
- the **solution to the problem** (Our Guess the Number game in PictoBlox).



Why Do Designers Study Existing Products?

Imagine creating a game without looking at any games first.

You might:

- ✗ Miss useful features
- ✗ Repeat mistakes
- ✗ Forget what users enjoy

Designers learn from existing products before creating their own.



Let's Analyse a Product

<https://plays.org/math-wizard/>

Example Product: **Math Wizard**

Strengths

- Gives immediate feedback
- Simple interface,
- Easy to play
- Includes a timer that makes the game exciting

Function

Helps users practise addition skills through a timed quiz game.



Inspiring Feature

The game provides instant feedback after every answer.

Areas of Improvement

It only focuses on addition and has limited game features.

How I Will Use This Feature in My Solution?

I will make my program give immediate feedback so users know whether their answer or action is correct.



Summative Task Introduction

For Strand iii you will:

- ✓ Choose one existing product
- ✓ Identify its purpose
- ✓ Describe its strengths
- ✓ Identify improvements
- ✓ Select one inspiring feature
- ✓ Explain how you will use it



Students may use websites, apps, games, teacher-provided examples, observations, screenshots or videos as sources for their product analysis.

Criteria	Example	Product 1
Product Name	Math Wizard	
Image /Link	https://plays.org/math-wizard/	
Purpose / Function	Helps users practise addition skills through a timed quiz game.	
Strengths	• Gives immediate feedback • Simple interface, • Easy to play • Includes a timer that makes the game exciting	
Areas of Improvement	It only focuses on addition and has limited game features.	
Inspiring feature	The game provides instant feedback after every answer.	
How I Will Use This Feature in My Solution?	I will make my program give immediate feedback so users know whether their answer or action is correct.	



Strand iv: Present the main findings of the relevant research

Command Term: Present **Present**

- Offer information clearly for others to understand.

Examples:

- Paragraph
- Poster
- Table
- Presentation



Strand iv: Present the main findings of the relevant research

Today we will:

- ✓ Review our research
- ✓ Identify important findings
- ✓ Summarize what we learned
- ✓ Use our findings to improve our solution

In this task: You will present your findings in a short paragraph.



What Are Research Findings?

Research findings are the important things you learned.

Example:

- **Research Question:**
What makes a game interesting?
- **Finding:**
Games are more engaging when they include feedback, scores and clear instructions.



Where Do Our Findings Come From?

- Our findings come from Strand ii, Strand iii

Strand ii: Research Planning

- User needs
- Coding blocks
- Features
- Interactive design

Strand iii: Product Analysis

- Existing products
- Strengths
- Improvements
- Inspiring features



What Should We Include?

Your paragraph should include:

- ✓ What users expect from an interactive program
- ✓ Features that make a solution interesting
- ✓ Useful ideas from existing products
- ✓ Coding techniques that may help
- ✓ Information that will help create your solution



Summative Task Introduction

Write a paragraph of approximately 50–100 words.

- Use information from:
- Your research points
- Your product analysis
- Your findings

to explain what you learned.



Success Checklist

Before submitting:

- ✓ I included information from my research.
- ✓ I included information from my product analysis.
- ✓ I explained useful features.
- ✓ I included coding ideas.
- ✓ I wrote a paragraph between 50–100 words.



Research Findings Summary: Using your research and product analysis, summarize the most important information you learned.

Include:

- What users expect from an interactive digital game or program.
- Features that make a digital solution interesting and easy to use.
- Useful ideas you discovered from existing products.
- Coding techniques that may help your solution work successfully.
- Key information that will help you create your interactive digital solution.

Write a paragraph of approximately 50–100 words.

